Oliver Colombini-Ljungberg

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Education

EPFL • Lausanne, Switzerland — 2022 – 2025
MSc in Robotics • Grade : 4.89/6

- Courses : Automatic Control (Model Predictive Control, Multivariable Control, System Identification), AI (Machine Learning, Deep Learning, Image Processing), Robotics (Mobile Robotics, Aerial Robotics, Legged Robots)
- Minor : Space Technologies (Finite Element Methods, Space Mechanisms, Earth Observation, System Engineering)

EPFL • Lausanne, Switzerland — 2019 – 2022
BS in Microengineering • Grade : 4.55/6

- Courses : Mechanics, Electronics, Mathematics, Computer Science

Research Experience

Ph.D. Candidate – Mines Paris – PSL, THALES — Sep 2025 –
Paris, France

- *Robotics, Operational Research, Perception :* Dynamic and collaborative planification for drone swarms. Multi-agent task assignment in uncertain conditions. Environment exploration, modeling and reconstruction. Perception, identification and tracking of targets. Grouped navigation under communication hazards
  Supervisors : Pr. Brigitte d'Andrea-Novel (Mines Paris – PSL), Pr. Pascal Morin (Mines Paris – PSL)

Master's thesis – ALTEN — Sep 2024 – Mar 2025
Toulouse, France

- *Robotics, Operational Research :* Optimising task assignment, navigation and path planning algorithms in the context of a fleet of Autonomous Mobile Robots that manage the logistics flows in a factory
  Supervisors : Pr. Auke Ijspeert (EPFL), Dr. Marcela Rojas Castro (ALTEN)

Minor's Project – EPFL — Feb 2024 – Jul 2024
Lausanne, Switzerland

- *Automatic Control, Machine Learning :* Use of Machine Learning methods to predict the behaviour of the actuators on a legged robot in order to improve the performances of its main controller
  Supervisor : Pr. Auke Ijspeert

Semester Project – EPFL — Sep 2023 – Feb 2024
Lausanne, Switzerland

- *Automatic Control, Deep Learning :* Use of Neural-ODE to improve the kinematic model and control of a mini-racecar
  Supervisor : Pr. Giancarlo Ferrari Trecate

Research Assistant – LiU — Jun 2023 – Sep 2023
Linköping, Sweden

- *Automatic Control :* Use of Nonlinear MPC to realize acrobatic stunts with a Crazyflie drone
  Supervisor : Pr. Daniel Axehill

Teaching Experience

Teaching Assistant – EPFL — 2021 – 2024
Lausanne, Switzerland

- *Numerical Analysis and Optimisation* for Pr. Marco Picasso
- *Analog Circuits and Systems* for Pr. Christian Enz

Tutoring – Acadomia, Anacours, Superprof — 2019 – 2025
Annecy, France

- Tutoring bachelor and high shool students in Mathematics, Phyics, Chemistry, Biology and Geology

Associations

Xplore – MAKE association from EPFL — 2023 – 2024
Lausanne, Switzerland

- *Robotics :* Design and construction of a rover and a legged robot for space exploration

Skills

- Programming : Python, C++/C, Matlab/Simulink, ROS, Gazebo, Catia, Assembler, Github/Gitlab, LaTeX
- Languages : French (native), Swedish (native), English (fluent), Spanish (intermediate)

References

- **Pr. Brigitte d'Andrea-Novel** • Full Professor – Mines Paris – PSL
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- **Pr. Pascal Morin** • Full Professor – Mines Paris – PSL
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